

# Mathematical illustration through a combinatorial lens

Christopher Hanusa  
Queens College, CUNY

[hanusa.xyz](https://hanusa.xyz)

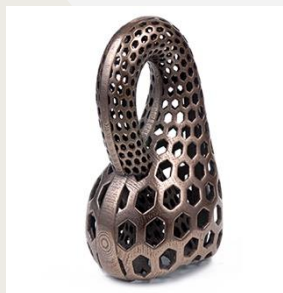
[@mathzorro](https://twitter.com/mathzorro)

# What makes a mathematical artist?

Mathematically infused artwork

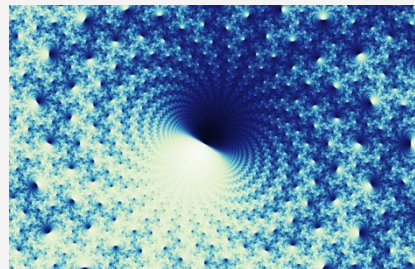
Illustrating mathematical concepts

## Bathsheba Sculpture



- Repetition
- Symmetry
- Geometry
- Higher Dimensions

## Henry Segerman

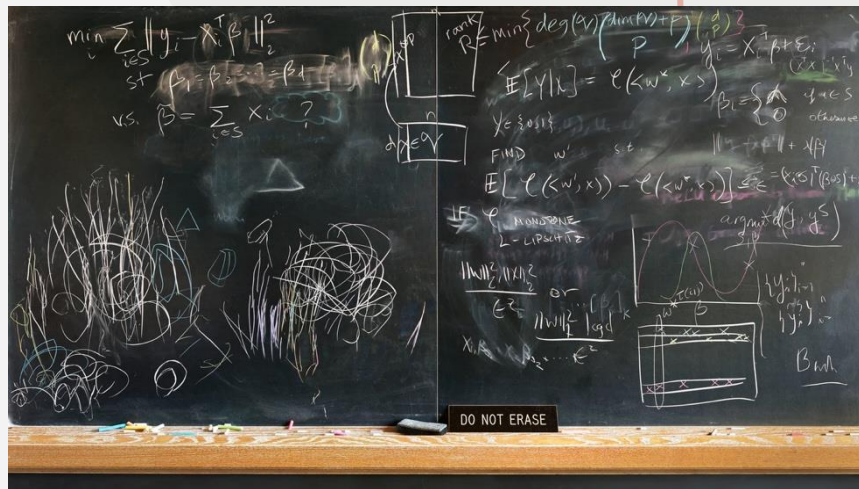


# My interest in Mathematical Art

Jessica Wynne, *Do Not Erase*  
(Laura Balzano U Mich 2019)

## Inherent to being a mathematician

- Desire to convey information precisely
- Mathematician's connection to their craft
- Physical manifestation of 3D concepts
- Outreach to the greater community
- Creative outlet



# Fundamental to **who I am** a combinatorist

**Bijections** between combinatorial interpretations

: “aw” them all!

**Domino Tilings:** Underlying Graph Structure

**Coxeter groups:** Alcoves, abacus diagrams (Zometools)

**Nonattacking chess pieces:** Hyperplane arrangements, discrete Fibonacci spiral

**Flow polytopes:** Structure of triangulations, flows on graphs, groves

**Best way to get me excited about a project:** Draw pictures and show patterns

# Becoming a Mathematical Artist

## Timeline

- 2014** I want to learn about 3D printing!
- 2015-16** Redesigned ***Mathematical Computing***
- 2016-17** Sabbatical
- 2017** Construct3D Conferences (also 18, 20)
- 2017** First Bridges MathArt Conference  
(Also 18, 22, 24, 25 + JMM)
- 2017-19** Connection with the SciArt Center
- 2017** Founded Hanusa Design

## Philosophically

- When is someone an artist?
- What makes an artist?
- Do I have something to say?
- Why should someone listen to me?
- Being able to ask questions without answers.

# First Bridges Conference

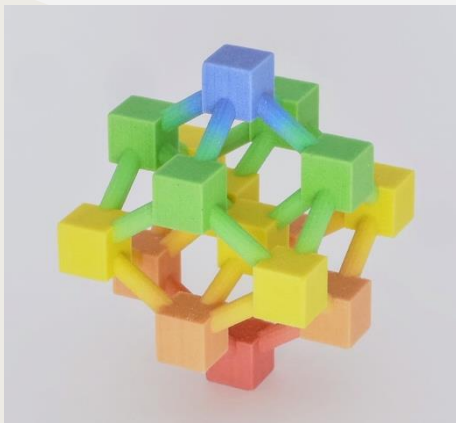
Waterloo, Ontario

Two pieces accepted (I brought more)

Accepted into a (math)art community

Brush with  
Greatness

Hasse Diagram



# My first show as an artist



SciArt Center (then SciArt Initiative)

Answer a "Call for Artwork"

External confirmation of interest  
in my work



# Riemannscapes

The New Normal (2017)

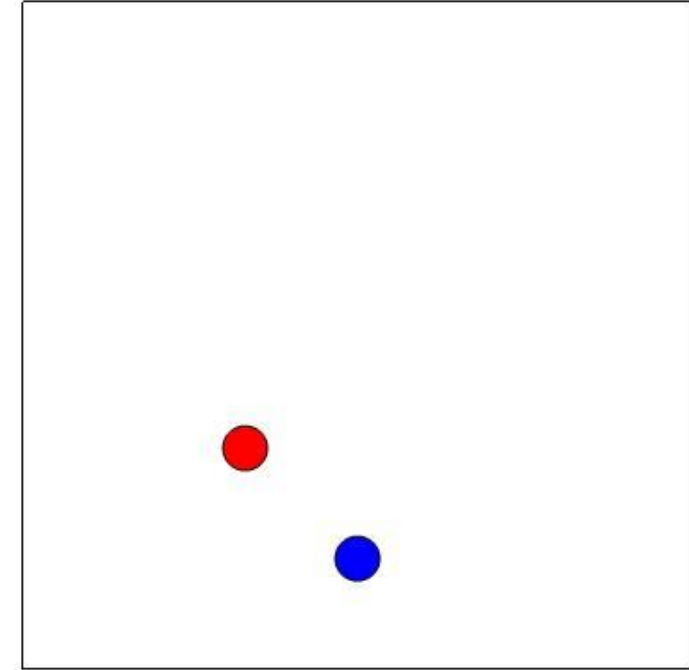
*Generative Art*



Create a random Voronoi Diagram on a disk

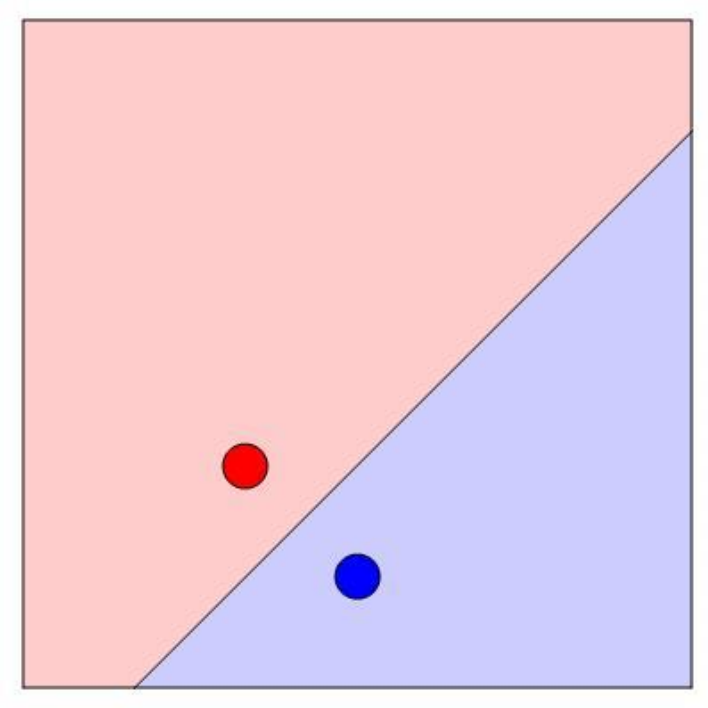
# Voronoi Diagrams

The **Voronoi Diagram**  
for a set of points is the **division of**  
**the region**  
into pieces  
based on **closeness**.



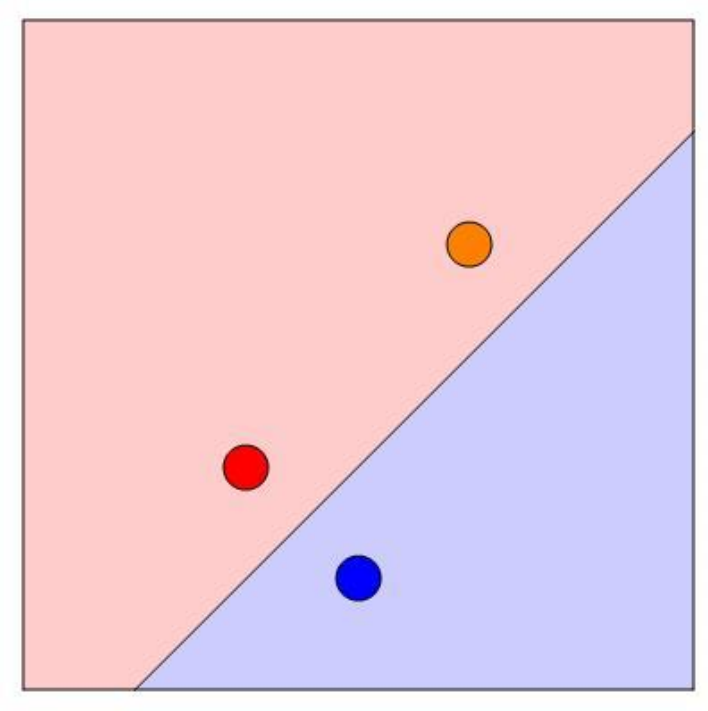
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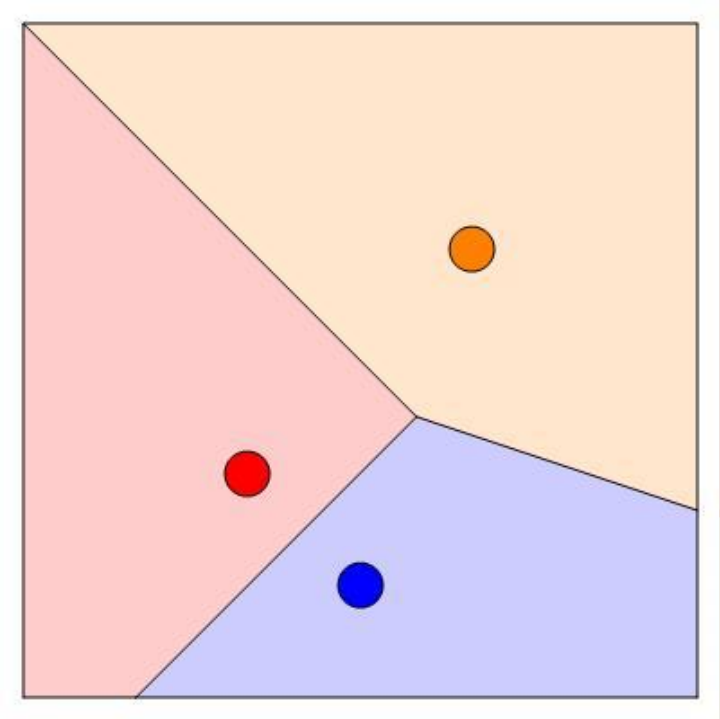
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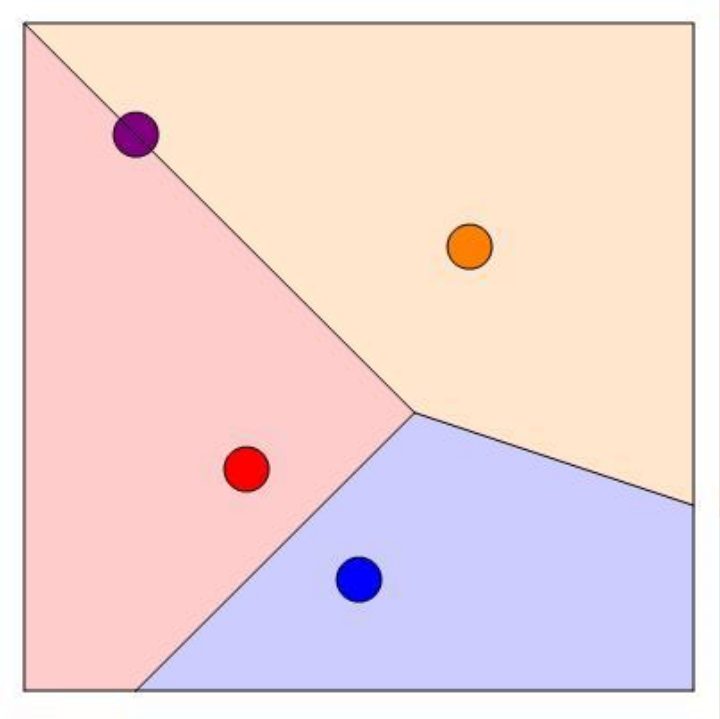
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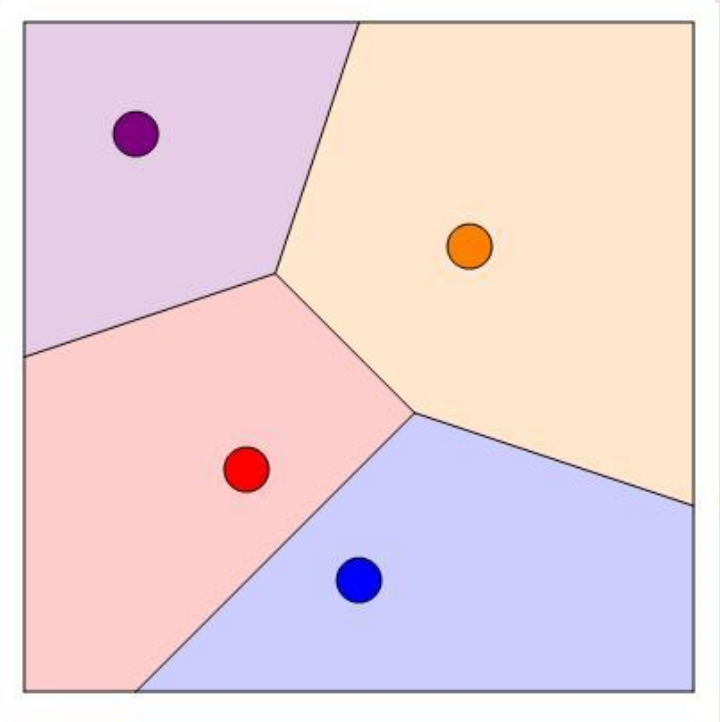
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# Riemannscapes

## The New Normal (2017)

### *Generative Art*



Create a random Voronoi Diagram on a disk

Choose a large subset of the cells

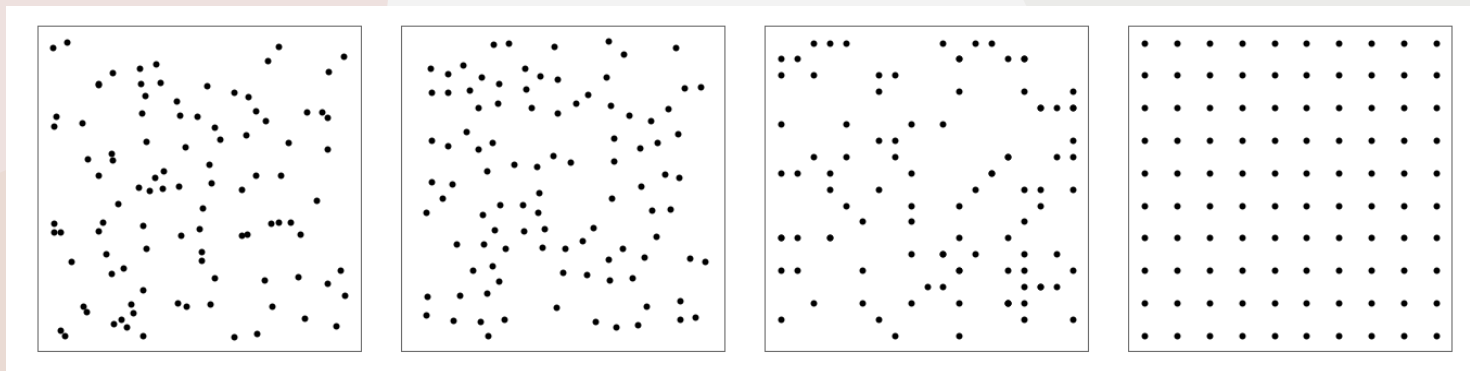
Raise each cell to the height of the  
2D Normal Distribution

Procedurally color the cells

**Question:** What does “random” mean?

# Voronoi Art

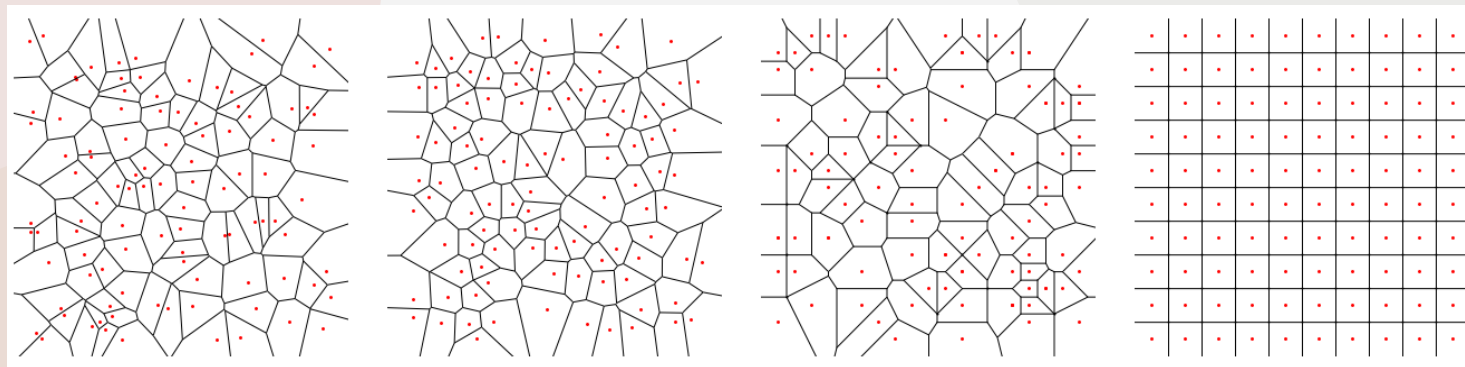
These points were generated  
using different randomization techniques.



Which is the most random?  
Which is the most beautiful?

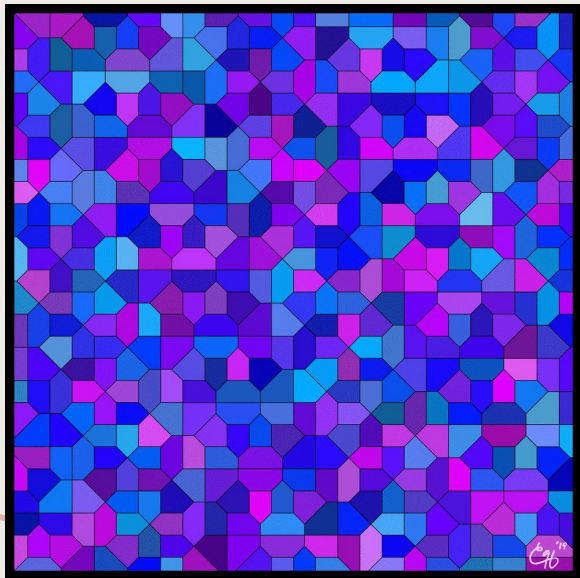
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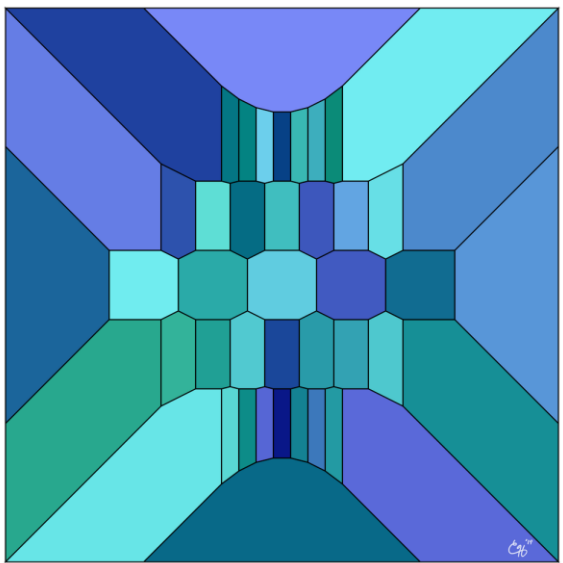


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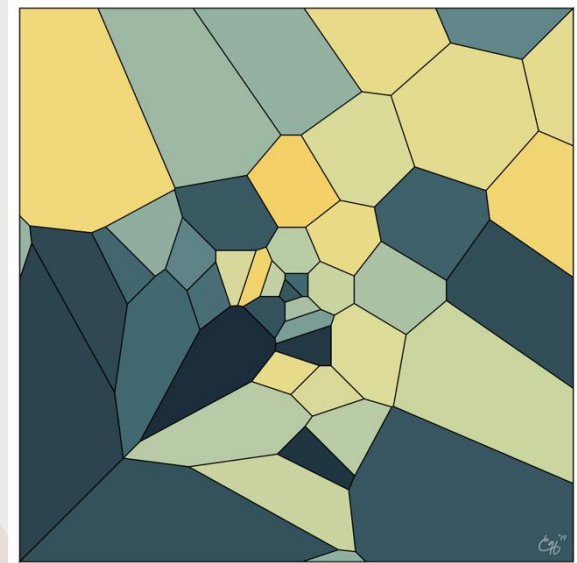
# Voronoi Art (2019)



Stained Glass



Pacific Northwest

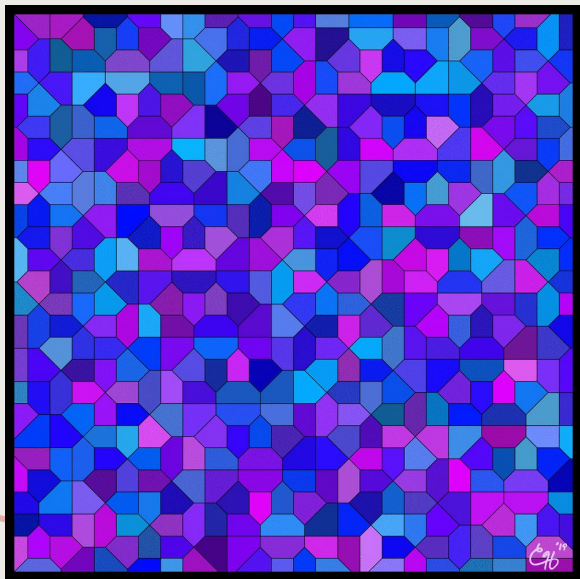


Lobster

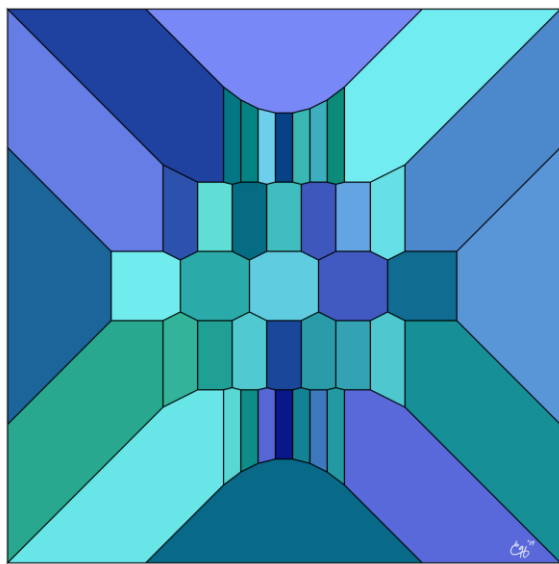
The background features several large, overlapping organic shapes in muted colors: light beige, pale pink, and soft grey. The shapes are layered, creating a sense of depth. The text is centered over these shapes.

**COLOR  
IS  
HARD**

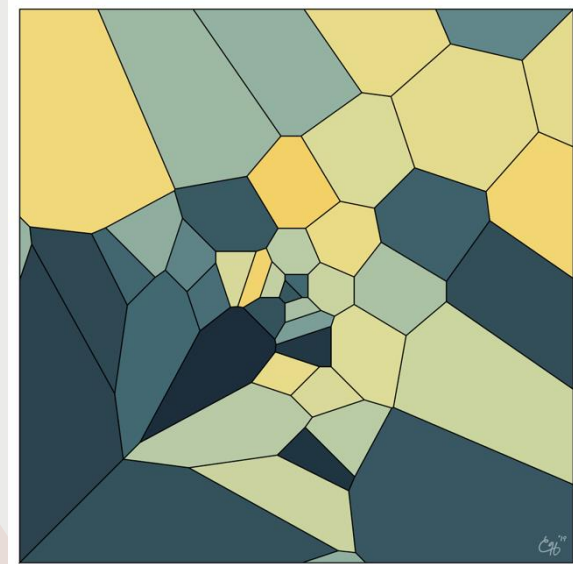
# Voronoi Art (2019)



Stained Glass



Pacific Northwest



Lobster

# Recurring theme: **Randomness** vs. **Deliberation**

At each step of the process,  
randomness informs the final piece.

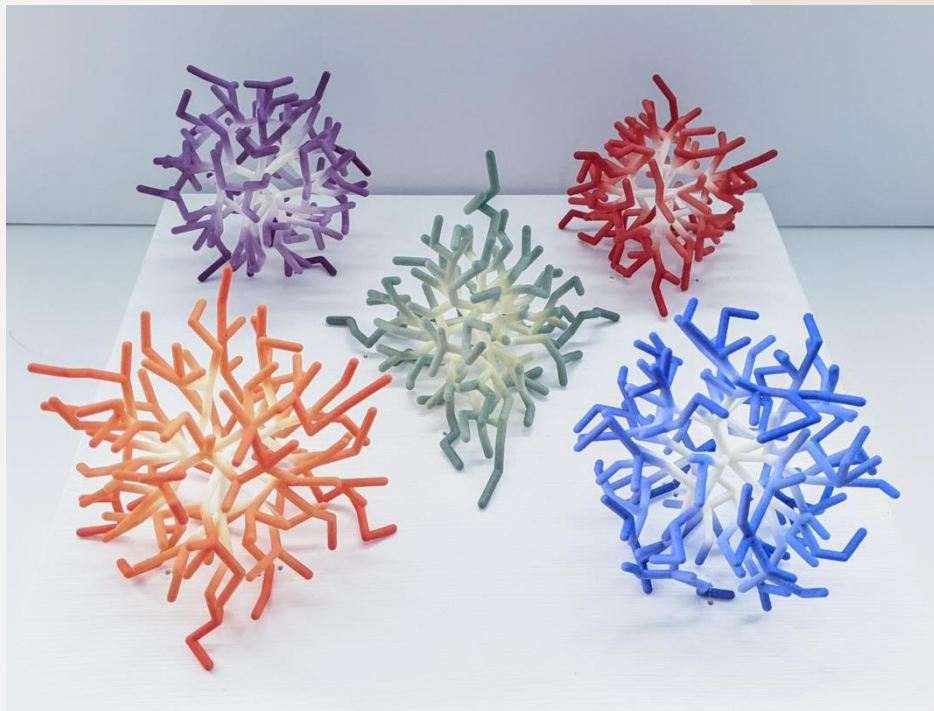
Once it yields something “good”,  
fix that aspect and develop further.

# Platonic Neurons

One for each platonic solid

- Choose random points in the solid
- Weight the distances
- Use Kruskal's algorithm
- Get min-weight spanning tree

3D printed in nylon then carefully hand dyed to create a radial gradient.



# Wolfram Mathematica is my medium

Switched in 2008; Research ... Visualization Capabilities

Inspires what is possible – there's a new feature.

What can I do with it? What can I learn?

**Interplay between teaching and creativity**

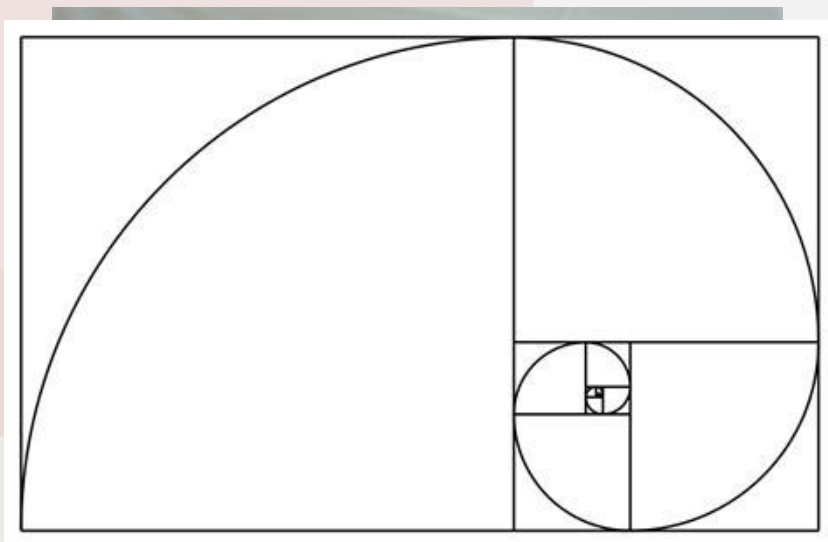
Mathematical Computing course: Teach students Mathematica.

Wanted to learn 3D printing.

So: Teach 3D printing! (and 3D modeling)

# Padovan Mountains

Many people know about the Fibonacci spiral.  
But do you know about the Padovan spiral?



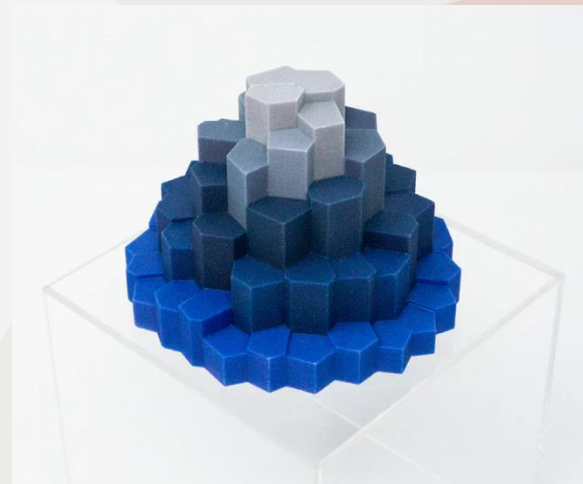
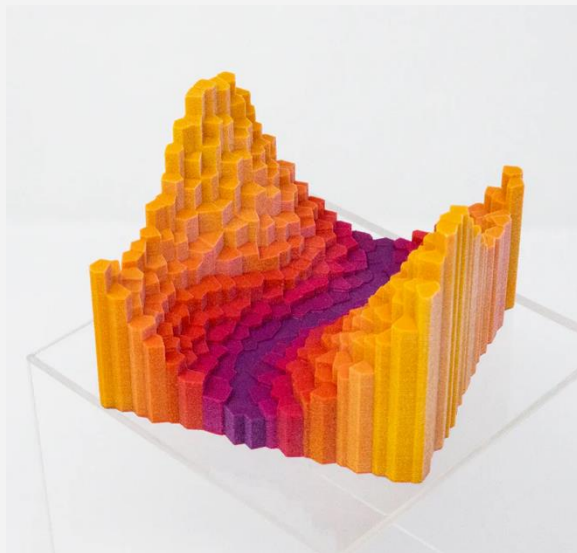


# SOLO SHOW

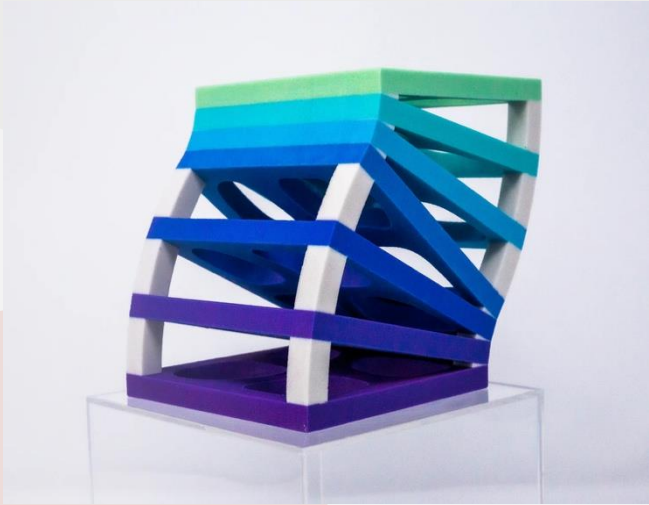
## *Existence and Uniqueness*

At Gallery 1064  
March 2022 • (Seattle, WA)

# Riemannscapes



# Window Evolution



Abstract forms change shape and size as you pass from window to window.  
You can see each individual frame by looking in from the sides;  
by looking from one side, you can recreate the entire form in your mind

# Recurring theme:

## Sight Lines in Three Dimensions

A wall in a 2D drawing is a line.  
You can see both sides.

A wall in a 3D sculpture is a wall.  
You cannot see what is behind it.

# Voronoi Chambers

Create a Voronoi diagram in 3D!

Choose equally spaced points  
on the unit sphere plus the origin.

Remove the walls of the central chamber  
to see inside.



# Becoming a Mathematical Artist

## Timeline

|                |                                                                 |
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| <b>2017-19</b> | Connection with the SciArt Center                               |
| <b>2017</b>    | Founded Hanusa Design                                           |
| <b>2022</b>    | Solo Show at Gallery 1064                                       |
| <b>2024</b>    | MADE@QC Gallery Curation                                        |
| <b>2024-25</b> | Interactive Pieces                                              |

## Philosophically

- When is someone an artist?
- What makes an artist?
- Do I have something to say?
- Why should someone listen to me?
- Being able to ask questions without answers.

# Intentionality

I wanted to share the ethos of mathematical art with non-majors

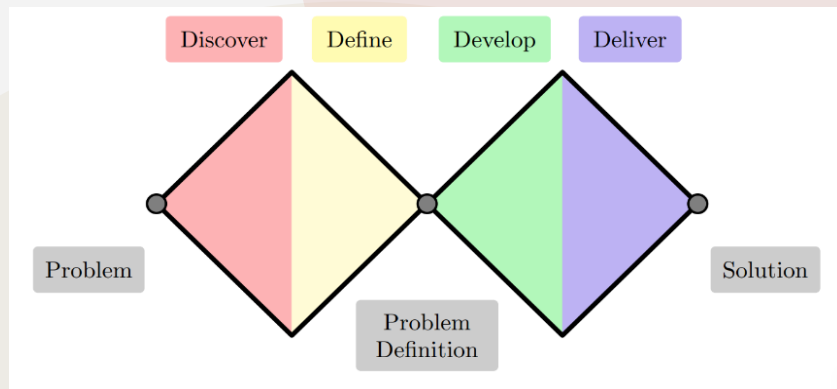
I created **Mathematical Design**.

How do you ...

- Teach how to make art?
- Infuse the Maker Mindset?
- Project-Based Learning
- Productive failure is OK and expected.

**Definition:** Make art using **intentionality**.

## Double Diamond Design Framework



# Becoming a Mathematical Artist

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## Philosophically

- When is someone an artist?
- **What makes art art?**
- Do I have something to say?
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# Five Signs You're an Artist





# Interactive Pieces

# Fractal Emergence

## Joint with Eric Vergo

Laser cut wood and acrylic  
LED Strips  
3D printed pieces  
Linear ball bearings  
Off the shelf Handles  
Weights



# Hyperbolic Emergence

Joint with  
Eric Vergo



# Experiential Braid Relations

The BEST way to understand permutations is with BRAIDS

- Multiplication
- Braid relations ( $s_1 s_2 s_1 = s_2 s_1 s_2$ )
- Non-commutativity

How to bring this to life?



# Recurring theme: Audience Engagement

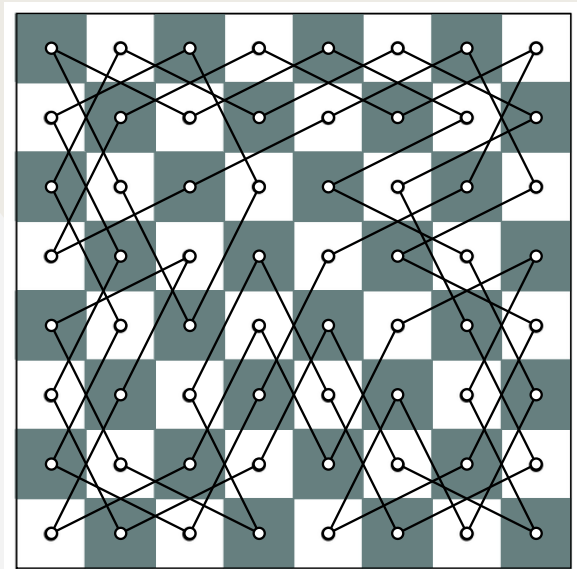
Active learning in the classroom engages  
students with the material

I'd love similar engagement  
with my artwork.

# Knight's Tours

Can a knight visit all the squares on an 8x8 chessboard?

Yes!

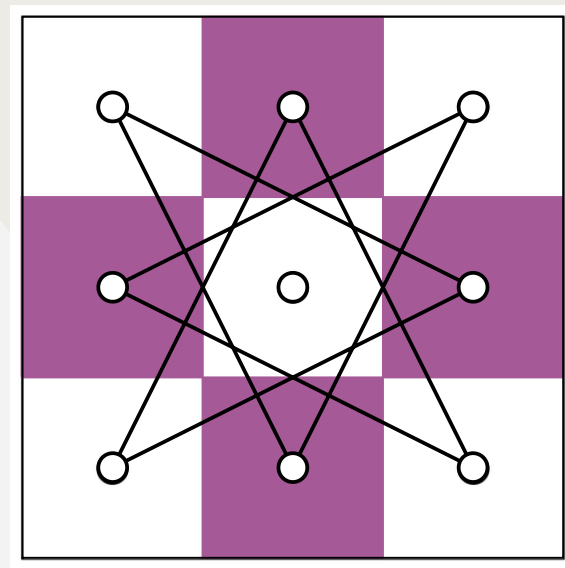


# 3x3 Knight's Tours

Can a knight visit all the squares on a 3x3 chessboard?

No!

Remove the central square.



# 3x3x3 Knight's Tours

Can a knight visit all the squares on a 3x3x3 chessboard?

No!

Remove the central cube.

Oops.



# 3x3x3x3 Knight's Tours

Central cell is "white" again.  
Remove to leave 80 cells.

There IS a Knight's Tour.  
Let's make it interesting.

No two consecutive moves are in the same plane.

How to visualize 4D?

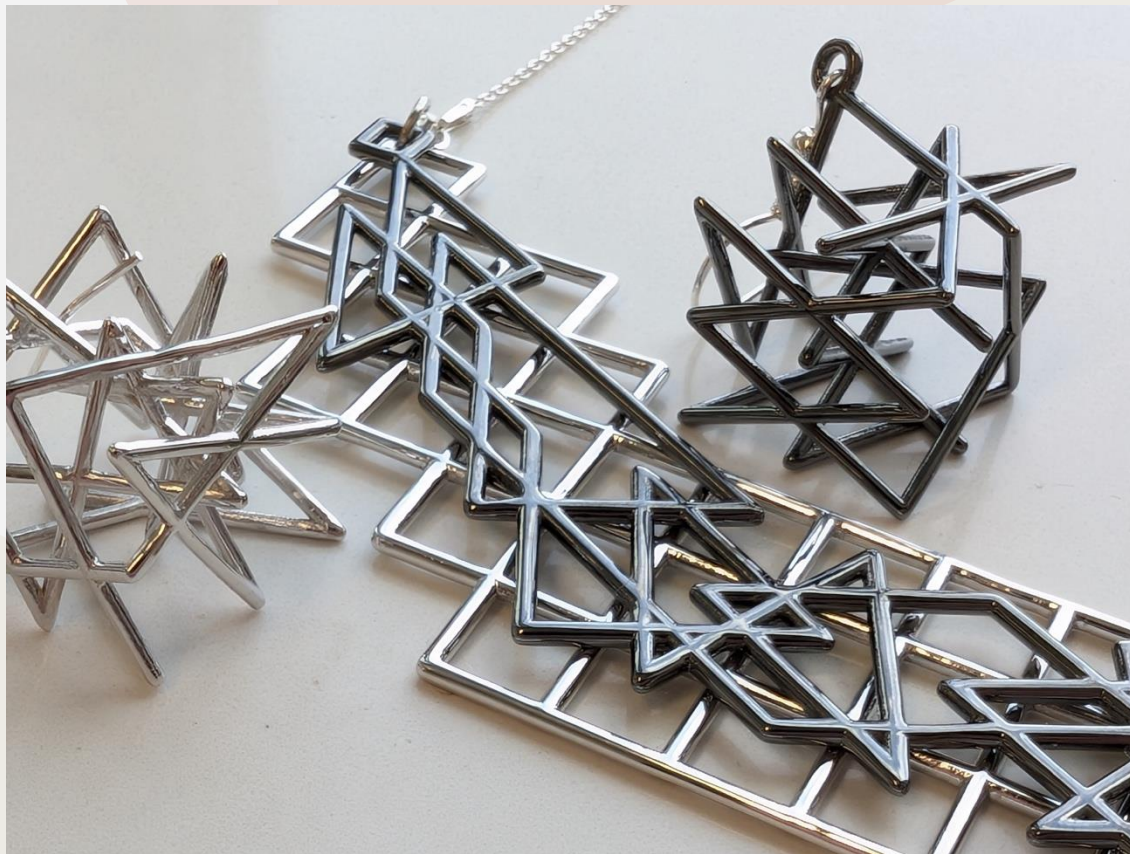
First three dimensions in a standard 3D grid. A unit move in the fourth dimension is represented by a translation of  $(1/4, 1/4, 1/4)$ ;

Same shaped vertices for a fixed 3D slice.



# Knight's Tour Jewelry

On the runway at the  
Bridges Fashion Show  
In Eindhoven in  
July 2025





# Hanusa Design

MATHEMATICAL JEWELRY

# Can I kick it? Yes I can!

Creating a business as a mathematician

Is it possible to create a jewelry business that could support me full time?

An existence question!

After a couple years, I have the answer:

Yeah, I think so.



# Small Business Crash Courses

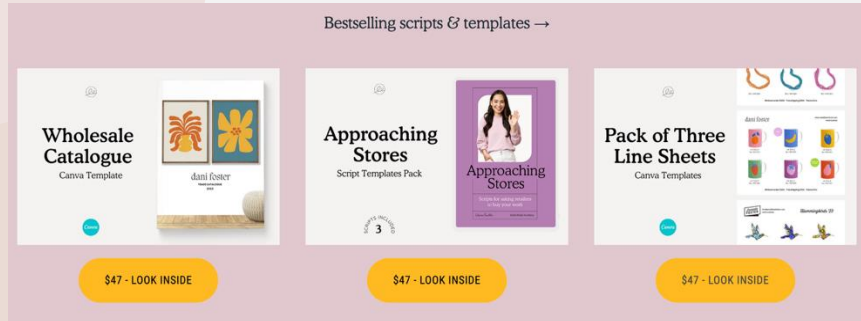
Indie Retail  
Academy  
Clare Yuille



You can sell  
your work to shops.

Welcome to Indie Retail Academy, where makers learn to sell  
their work to independent stores.

[READ ARTICLES](#)



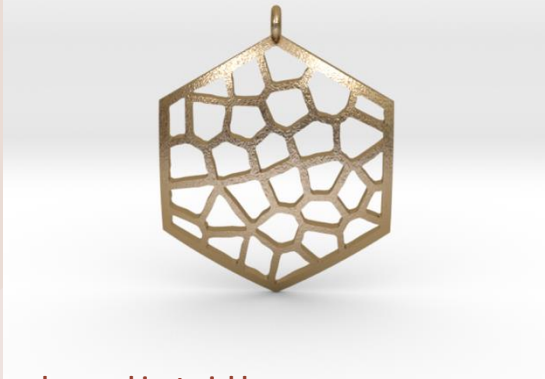
# Early Lessons

Exciting! Scary! What am I doing?!?!

- Initial idea: Steel pendants
- Print two copies of each in all steel types!



# Voronoi Jewelry



# Early Lessons

Exciting! Scary! What am I doing?!?!

- Initial idea: Steel pendants
- Print two copies of each in all steel types!
- **These are going to be amazing!**
- They are ... but they didn't sell.
- Lesson: Having a great idea doesn't make it a product that sells.
- What do people want to buy? How much are they willing to pay for it?



# Graph Theory

**What's the canonical graph in Graph Theory?**

What do we know about  $P$ ,  $K_{3,3}$ , and  $K_5$ ?



Forbidden Subgraph Earrings

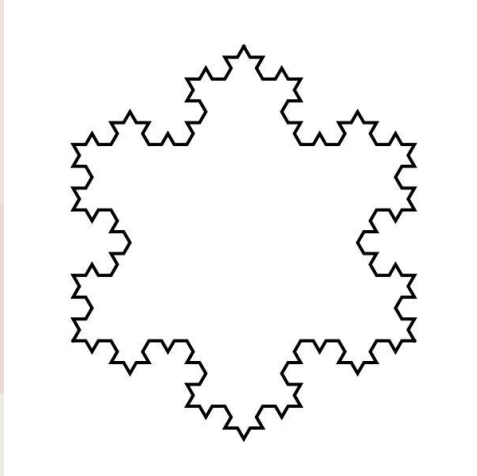


Petersen Graph Pendant .

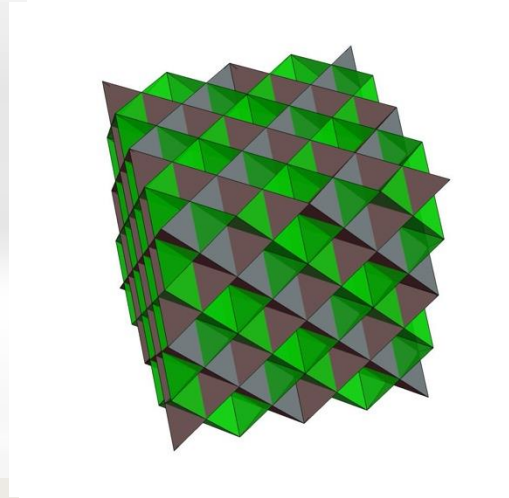
# Koch Fractals

Share beautiful concepts and learn something new

## Koch Snowflake



## Koch Tetrahedron



# Recurring theme:

## Generalize from 2D into 3D

Can we translate what we see in lower dimensions into higher dimensions?

Work to understand the underlying theory, implement it through algorithms, and hone the aesthetics to be appealing.

# Generalizing Hypercubes

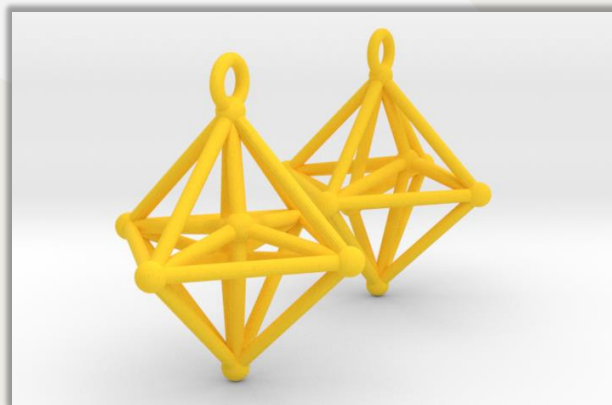
## Hypercube

Connect two copies at  
copied vertices



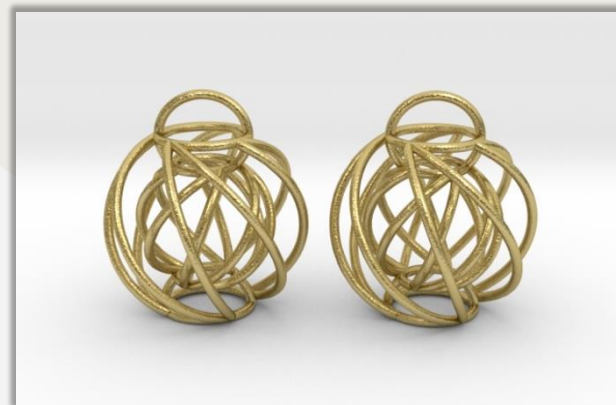
## Hyperoctahedron

Connect two vertices to  
every existing vertex



## Nested Spheres?

Create wireframes?!  
and see both spheres?!



# Born in this seminar

## Cobweb Plots (Katherine Moore)

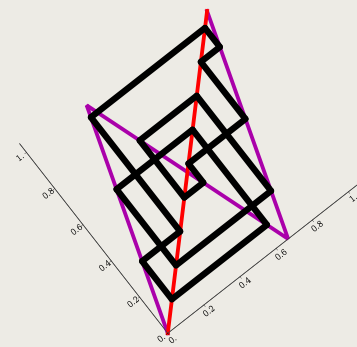
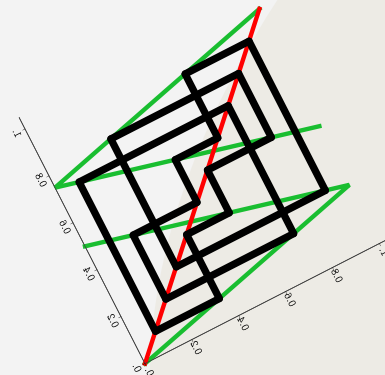
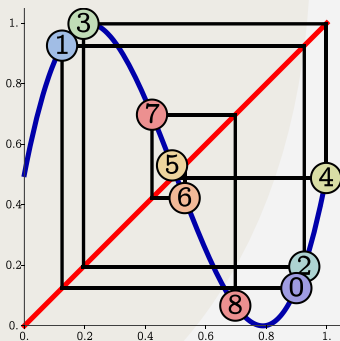
### Fall 2017 Talks

Sep 8, 2017: Kate Moore (Dartmouth College)

Title: Permutations in Time Series and Dynamical Systems

Abstract: Given a time series, we can extract a collection of permutations of a fixed length by sliding a window across our data and recording the relative order of the values in the window. It is known that distribution of permutations that appear conveys information about the complexity of the time series; one commonly studied measure is permutation entropy. Interestingly, in the specific case when a time series is defined by iterating a piecewise monotone map of the interval, there are forbidden permutations. Moreover, the number of allowed permutations encodes the system's topological entropy, a measure of complexity that is itself closely related to the permutation structure of periodic points.

Using permutation-based measures of entropy in time series analysis as motivation, I will talk about some interesting problems that arise at this intersection of combinatorics, dynamics and time series analysis.



# Cyclic Polytopes

A cyclic polytope is the convex hull of points on a helix.

It has the most number of edges possible for the number of vertices.



Thanks to Alejandro Morales

# Cipher

Limited edition collection





# Jewelry Sales

# Shops



QUEENS MUSEUM



Gallery North (L.I.)   Because Science (DC)   Wolfram Store (IL)



[hanusadesign.com](http://hanusadesign.com) (Shopify)

# Awesome-sauce

I've sold 2,500+ pieces of jewelry.

Outsized impact compared to my research articles.

Best feeling in the world...

I've been in fashion shows (!)

I've learned so much .....

# Bureaucra-sauce

- Which Suppliers?
- Cost / Quality / Responsiveness / Speed
- Tariffs
- Inventory
- Packaging
- Product photos, Product descriptions
- Self-promotion
- Social media
- Different ways to pay
- Mailing outside US
- Collecting Sales Tax (nexus?)
- Resale Certificates
- Filing Sales Tax
- Income tax: Money In, Cost of Goods Sold, Investments, Deductions
- Self-Employment Taxes
- Hiring?!?

The background features several large, overlapping organic shapes in muted colors: light gray, beige, and soft pink. These shapes are layered, creating a sense of depth. A thin, curved red line is visible in the bottom right corner.

End Matter

# Miscellaneous Projects

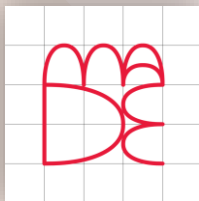
## Scarves



Collaboration with Lija  
Living (Philippines)

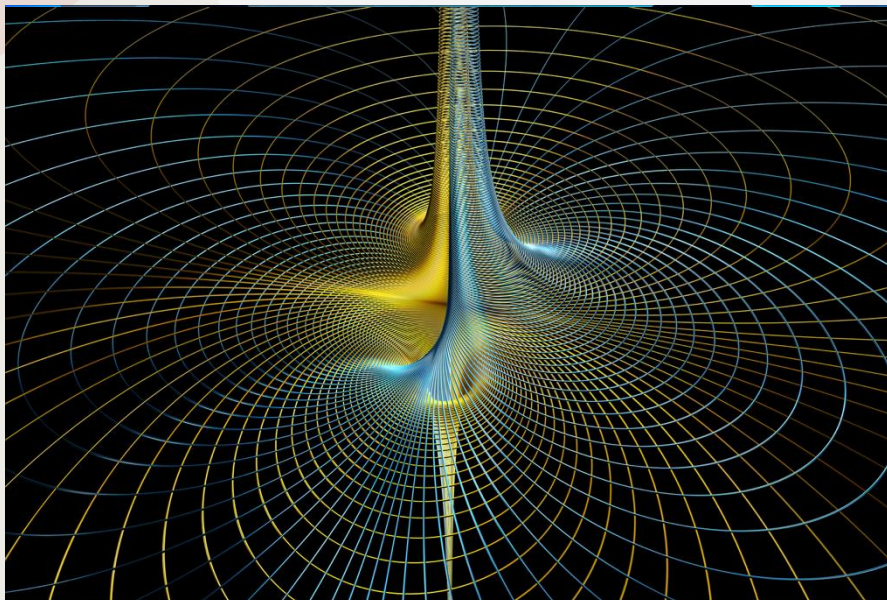
The background features several overlapping circles and curved lines in muted colors: a large light pink circle on the left, a light grey circle in the center, and a light beige circle on the right. Thin, curved lines in a reddish-pink color are also present, adding to the abstract design.

MADE@QC



# MADE@QC

Mathematical Art Digital Exhibition  
#mathart from around the world



## **Reality and Imagination (2023)**

Elke Stangl

*Custom javascript code in three.js*



Mathematical Art Digital Exhibition

### About the piece:

Real and imaginary part of the complex function  $1/z$ , envisaged as cityscape or space station. Celebrating the beauty of a seemingly simple function! The contour lines of these two functions are perfect circles - rings which are aligned along the circumference. Stacking these rings gives rise to tower-like structures. Gold is the color of reality; Blue is the color of imagination. The asymmetric towers lean on each other; they meet in the vertical axis, and they can be rotated into each other. In a series of images, these virtual structures are shown from different viewing angles.

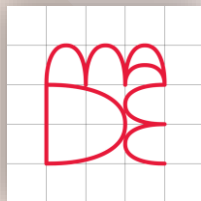
### Learn more about:

Complex  
Analysis



The artwork  
and artist

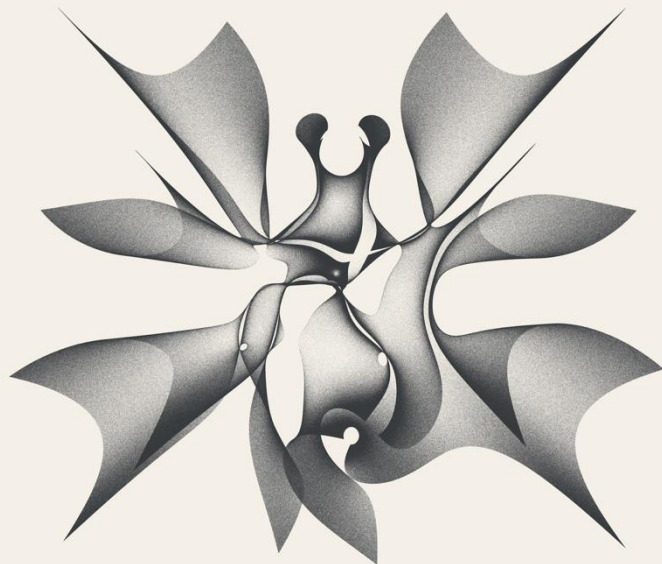




# MADE@QC

## Mathematical Art Digital Exhibition

### #mathart from around the world



### **Eigenfish (2024)**

Simone Conradi

Generated in Python using Matplotlib and Numpy packages

$$\begin{pmatrix} -i & -i & t_1 & -i & -i & 0 & -i & -i \\ i & 0 & 0.3 & 0 & 0.3 & -i & 0.3 & i \\ 0 & i & -i & i & 0 & 0.3 & i & 0 \\ -i & 0 & 0.3 & 0 & i & t_2 & 0.3 & 0 \\ 0 & 0.3 & 0.3 & -i & 0 & -i & 0 & 0.3 \\ -i & 0.3 & -i & i & 0 & -i & 0 & -i \\ 0 & 0.3 & -i & i & 0 & i & -i & 0 \\ -i & 0 & 0.3 & -i & 0.3 & i & 0 & i \end{pmatrix}$$



Mathematical Art Digital Exhibition

### About the piece:

Each artwork represents eigenvalues in  $\mathbb{C}$  for the shown parametric matrix, sampled with 2,000,000 values of the parameters  $t_1$  and  $t_2$  within the range  $[-20, 20] \times [-20, 20]$ . Each point on the graph is an eigenvalue in the complex plane. In the dark areas, the eigenvalues are denser, while in the light areas they are less dense. Dithering emerges from the natural distribution of eigenvalues.

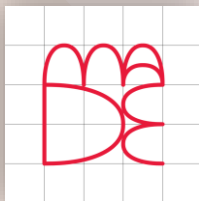
### Learn more about:

Eigenvalue  
Experiments



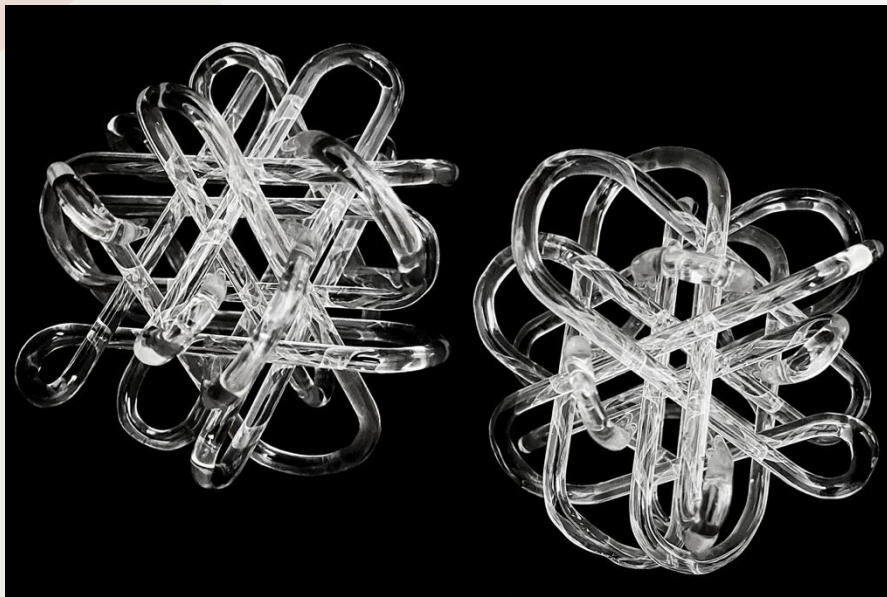
The artwork  
and artist





# MADE@QC

Mathematical Art Digital Exhibition  
#mathart from around the world



**Sigma Star 24 Stick Knot (2024)**

Anduriel Widmark

Flame worked Borosilicate Glass.



Mathematical Art Digital Exhibition

## About the piece:

The sculpture, "Sigma Star 24 Stick Knot", transforms theoretical math into tangible art by employing sigma star rod packing, a structure derived from systematically removing rods from a hexastix arrangement. This piece intricately loops and bends borosilicate glass rods into an embedded, symmetrical knot, representing one of the six invariant rod packings. The clarity of the borosilicate glass accentuates the sculpture's internal symmetry and structure, casting intricate shadows and reflections that reveal the underlying mathematical harmony.

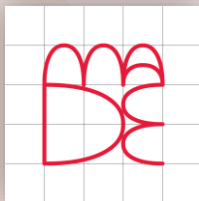
## Learn more about:

Hexastix



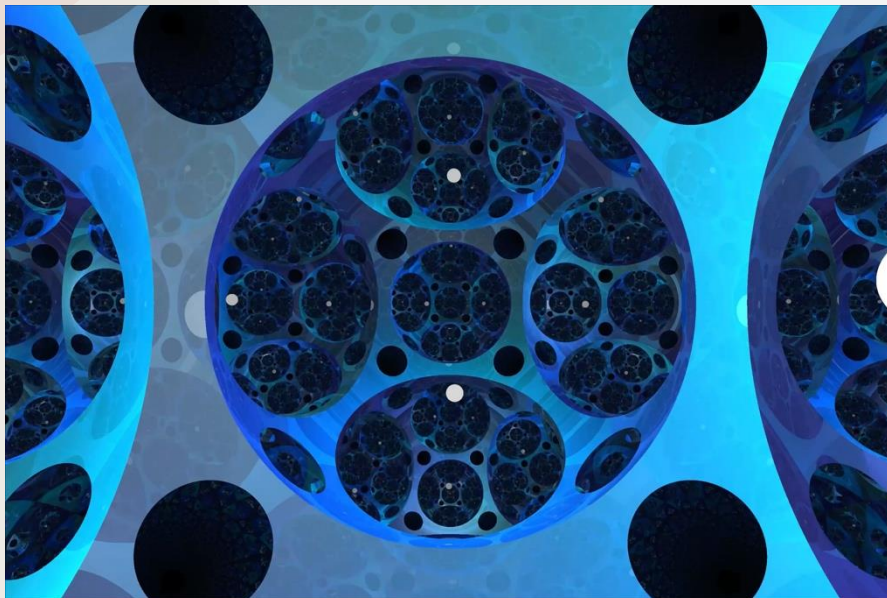
The artwork  
and artist





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## ***Inside Thurston geometries (2023)***

Rémi Coulon, Sabetta Matsumoto, Henry Segerman, and Steve Trettel  
*Digitally generated video. Software written in Javascript/GLSL.*

$$H^3 \quad \{(x, y, z, w) \in \mathbb{R}^4 \mid x^2 + y^2 + z^2 - w^2 = -1\}$$
$$ds^2 = dx^2 + dy^2 + dz^2 - dw^2$$

**Q** QUEENS  
COLLEGE

Mathematical Art Digital Exhibition

### About the piece:

3-dimensional.space is joint work by Rémi Coulon, Sabetta Matsumoto, Henry Segerman, and Steve Trettel to render accurate images of the eight Thurston geometries and their quotients. We make this software accessible to explore and encourage you to visit our website that hosts various demonstrations we have created along the way, as well as tutorials and documentation for the code.

### Learn more about:

Thurston  
Geometries



The artwork  
and artists



# Queens College Mathematical Discoveries Series

Lauren K. Williams

Wednesday, October 29, 2025

12:15-1:30pm

Talk for general audience; all are welcome.



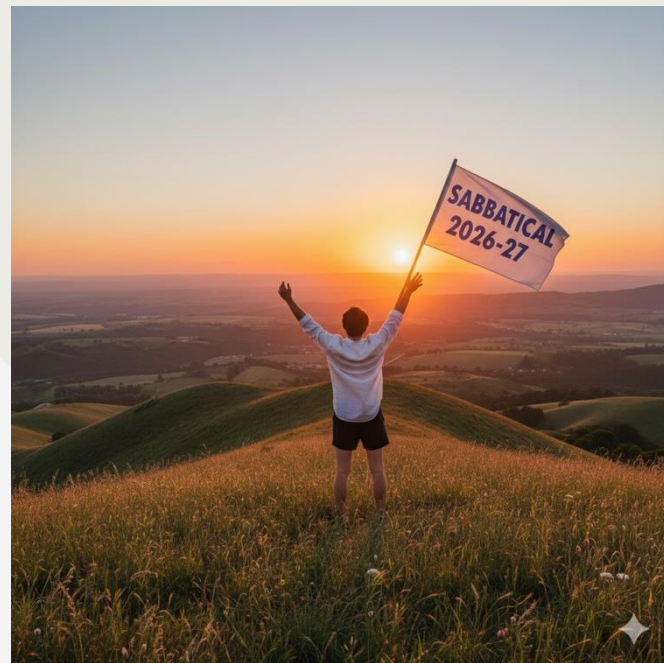
# How to get involved?

- Start creating.
- (By yourself) (With peers?) (With students?)
- False dichotomy: Quantity vs. Quality
- Share locally then globally.
- Join the Illustrating Mathematics Discord
- <https://discord.gg/ZJG8UgHX>



# What's next?

- How to represent families of combinatorial objects?
- Large scale artwork
- Looking for a partner for Hanusa Design
- More hats? Fewer hats?
- Finish up my Accreditation Responsibilities
- Sabbatical 2026-27!



# THANK YOU!

I am Christopher Hanusa



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> Research > Talks

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CREDITS: This presentation template was created by **Slidesgo**, and includes icons, infographics & images by **Freepik**



ART ✓OTE

The logo features the text "ART ✓OTE" in a bold, sans-serif font. The letters "ART" are magenta, the checkmark is green, and the letters "OTE" are green. The logo is centered within a white rectangular box. The background consists of large, overlapping, semi-transparent circles in shades of beige and light grey, with a thin, wavy red line in the lower right corner.

# Double Diamond Design Framework

**Definitions in language are as important as in Mathematics**

**What makes art art?**

**Infusing the maker mindset to my students – have to think about what making art is – intentionality -> further developing an artistic philosophy, learn about the Double Diamond Design Framework**

Mathematical Design – Gen Ed (Pathways) course for non-majors

**Project based learning**

**Failure is OK the classroom I integrate into my artwork and vice versa**